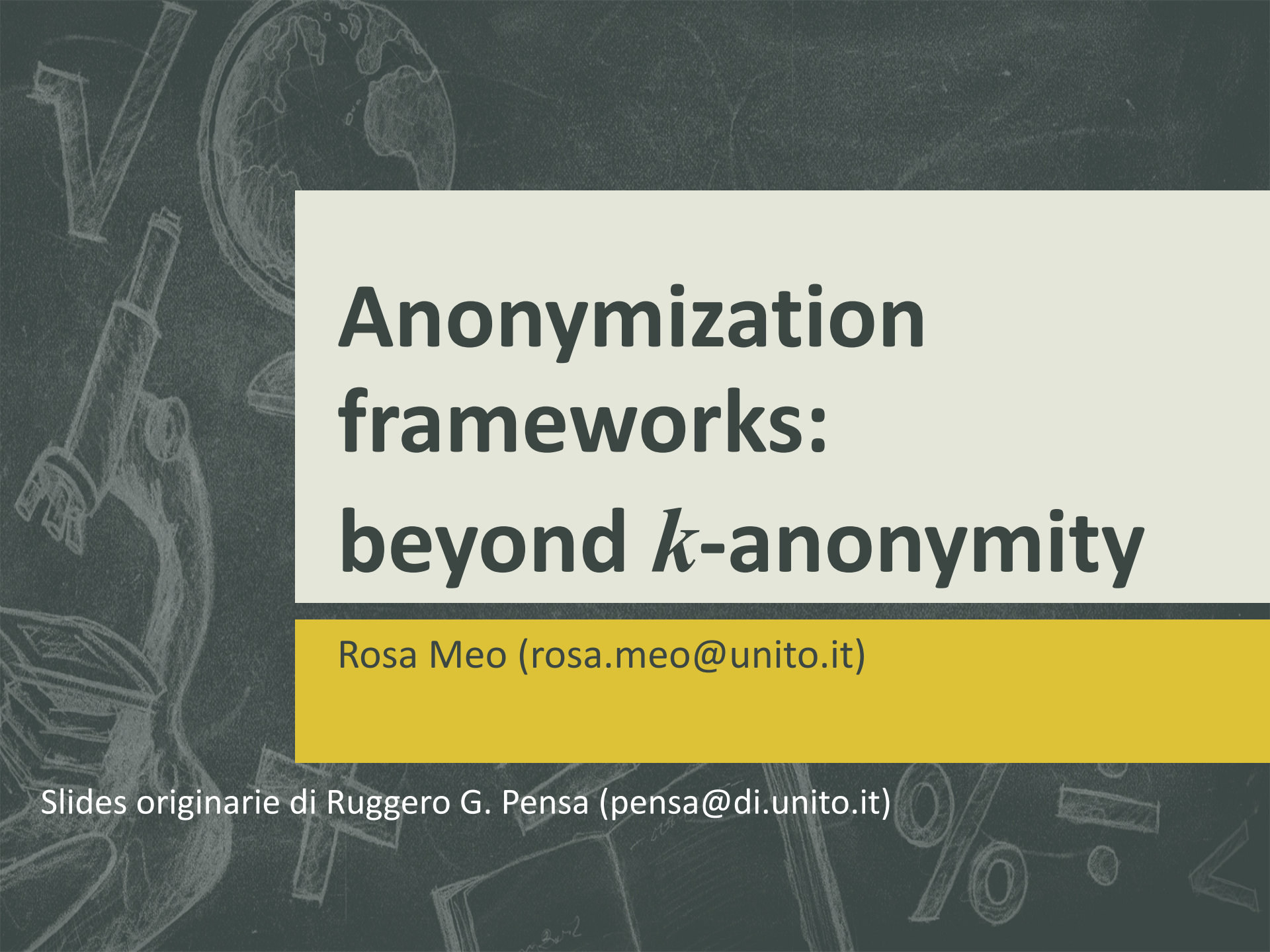




Anonymization frameworks: beyond k -anonymity

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Problems with k-anonymity

microdata

id	Zipcode	Age	National.	Disease
1	13053	28	Russian	Heart Disease
2	13068	29	American	Heart Disease
3	13068	21	Japanese	Viral Infection
4	13053	23	American	Viral Infection
5	14853	50	Indian	Cancer
6	14853	55	Russian	Heart Disease
7	14850	47	American	Viral Infection
8	14850	49	American	Viral Infection
9	13053	31	American	Cancer
10	13053	37	Indian	Cancer
11	13068	36	Japanese	Cancer
12	13068	35	American	Cancer

4-anonymous data

id	Zipcode	Age	National.	Disease
1	130**	<30	*	Heart Disease
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
5	1485*	≥40	*	Cancer
6	1485*	≥40	*	Heart Disease
7	1485*	≥40	*	Viral Infection
8	1485*	≥40	*	Viral Infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

Homogeneity attack

It may happen when all tuples with a quasi identifier value in a k-anonymity table have the same sensitive attribute value

- An attacker knows that Carol is an 31-years-old American woman living in 13053
- The attacker also known that Carol's data are in the microdata table
- The attacker can infer that Carol suffers of cancer

id	Zipcode	Sex	National.	Disease
...	...	...	...	...
9	13053	31	American	Cancer
10	13053	37	Indian	Cancer
11	13068	36	Japanese	Cancer
12	13068	35	American	Cancer

id	Zipcode	Sex	National.	Disease
...	...	...	...	...
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

Background knowledge attack

Based on a prior knowledge of some additional external information

- An attacker knows that Hellen is a 29-years-old American woman living in 13068
- The attacker can infer that the disease of Hellen is either Viral Infection or Heart Disease
- But the attacker also know that Hellen is a sportswoman!

id	Zipcode	Age	National.	Disease
1	13053	28	Russian	Heart Disease
2	13068	29	American	Heart Disease
3	13068	21	Japanese	Viral Infection
4	13053	23	American	Viral Infection
...	...	...	...	...

id	Zipcode	Age	National.	Disease
1	130**	<30	*	Heart Disease
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
...	...	...	...	...

Positive and negative disclosure

Given the adversary's background knowledge, a published table, generalized (GT) might disclose information in two important ways

- **positive disclosure:** publishing the table GT that was derived from a private table (PT) results in a positive disclosure if the adversary can correctly **identify** the value of a sensitive attribute with high probability
- **negative disclosure:** publishing the table GT that was derived from PT results in a negative disclosure if the adversary can correctly **eliminate** some possible values of the sensitive attribute (with high probability)

Negative disclosure (example)

Even without background knowledge, the attacker can deduce that Hellen (the 29-years-old American woman living in 13068) does not have cancer

id	Zipcode	Age	National.	Disease
1	13053	28	Russian	Heart Disease
2	13068	29	American	Heart Disease
3	13068	21	Japanese	Viral Infection
4	13053	23	American	Viral Infection
...	...	...	...	...

PT

id	Zipcode	Age	National.	Disease
1	130**	<30	*	Heart Disease
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
...	...	...	...	...

GT

Definitions

PT: private table

GT: generalized table

- Let q be a value of a nonsensitive attribute Q in PT
- Let q^* be the generalized value of q in GT
- Let s be a possible value of the sensitive attribute S
- Let $n(q^*, s)$ be the number of tuples $t^* \in GT$ where $t^*[Q] = q^*$ and $t^*[S] = s$
- Let a q^* -block (aka **equivalence class**) be the set of tuples in GT whose nonsensitive values of Q generalize to q^*

Lack of diversity

Lack of diversity in a sensitive attribute S occurs when

$$\forall s' \neq s \quad n(q^*, s') \ll n(q^*, s)$$

In this case, $\Pr(t[S]=s \mid t^*[Q]=q^*) \approx 1$ and the sensitive value s of the person identified by $t[Q]=q$ can be determined with high accuracy by the attacker, provided that she knows that the person is in PT

l -diversity (principle)

A. Machanavajjhala, J. Gehrke, D. Kifer, M. Venkatasubramanian: l -Diversity: Privacy Beyond k -Anonymity. ICDE 2006: 24

- To ensure **diversity** a q^* -block should have at least $l \geq 2$ different sensitive values such that the l most frequent values (in the q^* -block) have roughly the same frequency
- Such a q^* -block is **well-represented** by l **sensitive values**

l -diversity and background knowledge

- If there are l well represented sensitive values in a q^* -block, then the attacker needs $l - 1$ damaging pieces of background knowledge to eliminate $l - 1$ possible sensitive values and infer a positive disclosure
- By setting the parameter l , the data publisher can determine how much protection is provided against background knowledge (even if this background knowledge is unknown to the publisher)

l -diversity principle (global)

A q^* -block is l -diverse if contains at least l well-represented values for the sensitive attribute S

A table is l -diverse if every q^* -block is l -diverse.

Example: 3-diverse data

4-anonymous data

id	Zipcode	Age	National.	Disease
1	130**	<30	*	Heart Disease
2	130**	<30	*	Heart Disease
3	130**	<30	*	Viral Infection
4	130**	<30	*	Viral Infection
5	1485*	≥40	*	Cancer
6	1485*	≥40	*	Heart Disease
7	1485*	≥40	*	Viral Infection
8	1485*	≥40	*	Viral Infection
9	130**	3*	*	Cancer
10	130**	3*	*	Cancer
11	130**	3*	*	Cancer
12	130**	3*	*	Cancer

4-anonymous / 3-diverse data

id	Zipcode	Age	National.	Disease
1	1305*	≤40	*	Heart Disease
4	1305*	≤40	*	Viral Infection
9	1305*	≤40	*	Cancer
10	1305*	≤40	*	Cancer
5	1485*	>40	*	Cancer
6	1485*	>40	*	Heart Disease
7	1485*	>40	*	Viral Infection
8	1485*	>40	*	Viral Infection
2	1306*	≤40	*	Heart Disease
3	1306*	≤40	*	Viral Infection
11	1306*	≤40	*	Cancer
12	1306*	≤40	*	Cancer

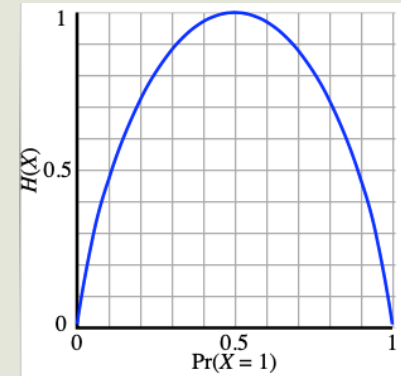
Definition: entropy l -diversity

A table GT is **entropy l -diverse** if, for every q^* -block

$$-\sum_{s \in S} p(q^*, s) \log(p(q^*, s)) \geq \log(l)$$

where $p(q^*, s) = \frac{n(q^*, s)}{\sum_{s' \in S} n(q^*, s')}$ is the fraction of tuples in

the q^* -block with sensitive attribute value equal to s



Theorem: monotonicity of l -diversity

Entropy l -diversity satisfies the **monotonicity** property: if a table GT satisfies entropy l -diversity, then any generalization of GT also satisfies entropy l -diversity

- **Consequence:** any algorithm for k -anonymity can be extended to enforce the l -diversity property

Attacks on l -diversity

l -diversity leaves space to attacks based on the distribution of values inside q^* -blocks (equivalence classes)

- **skewness attack:** happens when the distribution in a q -block is different than the original population
- **similarity attack:** happens when a q -block has different but semantically similar values for the sensitive attribute

About membership disclosure

- With large probability, quasi-identifier is unique in the population
- But generalizing/suppressing quasi-identifiers in the dataset does not affect their distribution in the population (obviously!)
 - Suppose anonymized dataset contains 10 records with a certain quasi-identifier ...
... and there are 10 people in the population who match this quasi-identifier
- k-anonymity **may not hide** whether a given person is in the dataset

Presence of individual in data

- Simpler but practical problem: the risk is simply **from identifying that an individual is (or is not) in an anonymized dataset**
- Necessity to publish a dataset to support research on a specific condition, but identifying individuals meeting that condition is damaging
- Examples
 - **counter-terrorism**: database containing information about suspected terrorist groups to support research in automated support for discovering terrorism
 - **medical research**: database of patients with a particular type of cancer

δ -presence

M. E. Nergiz, M. Atzori, C. Clifton: Hiding the presence of individuals from shared databases. SIGMOD Conference 2007: 665-676

Given an external public table T , and a private table PT , we say that **δ -presence holds for a generalization GT** of PT , with $\delta = (\delta_{\min}, \delta_{\max})$ if

$$\delta_{\min} \leq P(t \in PT \mid GT) \leq \delta_{\max} \quad \forall t \in T$$

In such a dataset, we say that each tuple $t \in T$ is δ -present in PT

$\delta = (\delta_{\min}, \delta_{\max})$ is a range of acceptable probabilities for $P(t \in PT \mid GT)$

Example

Public table T

ID	ZIP	Age	Nationality	Sens.
a	47906	35	USA	0
b	47903	59	Canada	1
c	47906	42	USA	1
d	47630	18	Brazil	0
e	47630	22	Brazil	0
f	47633	63	Peru	1
g	48973	33	Spain	0
h	48972	47	Bulgaria	1
i	48970	52	France	1

publicly known data

Research subset PT (with quasi-id)
with patients with sensitive $s=1$

ID	Zip	Age	Nationality
b	47903	59	Canada
c	47906	42	USA
f	47633	63	Peru
h	48972	47	Bulgaria
i	48970	52	France

Example

Generalized table GT

	ZIP	Age	Nationality	Sens.
a	47*	*	America	0
b	47*	*	America	1
c	47*	*	America	1
d	47*	*	America	0
e	47*	*	America	0
f	47*	*	America	1
g	48*	*	Europe	0
h	48*	*	Europe	1
i	48*	*	Europe	1

publicly released dataset

$$P(a \in PT | GT) = \frac{|\{b, c, f\}|}{|\{a, b, c, d, e, f\}|} = \frac{1}{2}$$

the same for b, c, d, e and f

$$P(g \in PT | GT) = \frac{|\{h, i\}|}{|\{g, h, i\}|} = \frac{2}{3}$$

The same for h and i

Generalized research subset PT

	Zip	Age	Nationality
b	47*	*	America
c	47*	*	America
f	47*	*	America
h	48*	*	Europe
i	48*	*	Europe

Example

Generalized table GT

ID	ZIP	Age	Nationality	Sens.
a	47*	*	America	0
b	47*	*	America	1
c	47*	*	America	1
d	47*	*	America	0
e	47*	*	America	0
f	47*	*	America	1
g	48*	*	Europe	0
h	48*	*	Europe	1
i	48*	*	Europe	1

publicly released dataset

GT is a $\left(\frac{1}{2}, \frac{2}{3}\right)$ -generalization of PT

Generalized research subset PT

ID	Zip	Age	Nationality
b	47*	*	America
c	47*	*	America
f	47*	*	America
h	48*	*	Europe
i	48*	*	Europe

Monotonicity of δ -presence

Given a public table T , private table PT , a generalization GT^1 of PT , and a generalization GT^2 of GT^1 , if GT^1 is $(\delta_{\min}, \delta_{\max})$ -present w.r.t. T and PT then so is GT^2

In other words

If GT^2 is not δ -present w.r.t. T and PT then neither is GT^1

Nice property that can be exploited by most k -anonymity algorithms

Selecting a good δ

Example: release of a dataset related to diabetes patients

Let I_p be the event that person p has diabetes

Since the rate of diabetes in all US population is public information,
any adversary will have a prior belief b_r on I_p given the public dataset
 T

$$b_r = P(I_p) = 0.07$$

Selecting a good δ

The private dataset PT is a subset of diabetes patients.
 Seeing some anonymization GT of the table T, attacker will have a posterior belief b_o on I_p :

$$\begin{aligned}
 b_o &= P(I_p | GT) = \\
 b_o &= P(I_p | p \in PT) \cdot P(p \in PT | GT) + P(I_p | p \notin PT) \cdot P(p \notin PT | GT) \\
 &\quad \swarrow \text{=1 because PT contains only sick people} \quad \quad \quad \text{number of sick people in public table T} \quad \quad \quad \text{Number of sick people not in PT} \\
 &\quad \quad \quad b_r \quad \quad \quad \boxed{P(I_p) \cdot |T| - |PT|} \\
 &= 1 \cdot P(p \in PT | GT) + \frac{\boxed{P(I_p) \cdot |T| - |PT|}}{\boxed{|T| - |PT|}} \cdot [1 - P(p \in PT | GT)] \\
 &\quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \quad \text{Number of people not in PT} \\
 &= \underbrace{P(p \in PT | GT)}_{\delta} \cdot \frac{|T| \cdot (1 - b_r)}{|T| - |PT|} + \frac{b_r \cdot |T| - |PT|}{|T| - |PT|} \quad (\text{eq. 1})
 \end{aligned}$$

Selecting a good δ

Example:

We consider an acceptable cost due to misuse: assume a hiring decision, and that a \$100 annual difference in total cost of employee is noise (difference in productivity, taking an extra sick day, salary negotiation, etc.)

If expected annual cost of medical treatment of diabetes based on misuse of the database is $c < \$100$, the risk of misuse is acceptably small

The total cost of diabetes per person is around $d = \$10,000$

We must ensure

expected cost (risk) a posteriori

$$b_o \cdot d - b_r \cdot d < c$$

If $|PT| \cong 0.04 |T|$, we obtain

prior expected cost

proportion of costs or acceptable risk of misuse

$$b_o - b_r < \frac{c}{d} = \frac{1}{100} \Rightarrow b_o < \frac{1}{100} + b_r$$

gained belief after publishing research data
(difference between posterior belief and prior belief)

(substituting in eq (1) δ ; $0.04 \cdot |T|$ in $|PT|$; 0.07 in b_r ; $0.01 + b_r$ in b_o , gives the upper bound for δ):

$$\delta_{max} = P(p \in PT | GT) \sim 0.05$$

Selecting a good δ

From eq. 1, substituting the value of $b_r=0.07, |PT|=0.04*|T|$, eliminating $|T|$, we obtain:

$$b_0 = \delta \cdot \frac{1 - 0.07}{1 - 0.04} + \frac{0.07 - 0.04}{1 - 0.04} = \delta \cdot 0.97 + 0.03$$

We need to ensure that the company can't "cherry-pick" employees known **not to be** in the database

Thus the posterior belief b_0 should not be arbitrarily low for some persons p ;
it should be comparable to b_r  gives a lower bound of $\delta = \delta_{min}$

$$b_0 = b_r = 0.07 = \delta \cdot 0.97 + 0.03 \quad \text{} \quad \delta = \frac{0.07 - 0.03}{0.97} = 0.04$$

We obtain:

$$\forall p : P(p \in PT | GT) \geq 0.04 = \delta_{min}$$

Conclusion on k -anonymity approaches

- **Protection against utility measures:** research is needed to develop measures to allow users to assess, besides the protection offered by the data, the utility of the released data
- **Efficient algorithms:** computing a table that satisfies k -anonymity guaranteeing minimality is an NP-hard problem
- **New techniques:** the k -anonymity property is not tied to a specific technique

Conclusion on k -anonymity approaches

- **Merging of different tables and views:** the original k -anonymity proposal as well as most subsequent work assume
 - The existence of a single table to be released
 - The released table contains at most one tuple for each respondent
- **External knowledge:** k -anonymity did not model external knowledge that can be further exploited for inference and expose the data to identity or attribute disclosure
- and...

Curse of dimensionality

C. C. Aggarwal: On k-Anonymity and the Curse of Dimensionality. VLDB 2005: 901-909

- Generalization fundamentally relies on **spatial locality**
 - Each record must have k close neighbors
- Real-world datasets are very sparse
 - Many attributes (dimensions)
 - Netflix Prize dataset: 17,000 dimensions
 - Amazon customer records: several million dimensions
 - “Nearest neighbor” is very far
- Projection to low dimensions loses all info $\Rightarrow$
k-anonymized datasets are useless

